

Renfred Alonge

Mississauga, Canada | +1 (647) 457-6660 | alongerenfred@gmail.com | LinkedIn | GitHub

EDUCATION

University of Toronto – Mississauga

Expected Jun. 2029

B.Sc. Computer Science and Mathematics

- Lester B. Pearson International Scholar (2025 Cohort), GPA: 4.0/4.0
- Dean's List Scholar (Year 1)
- Relevant Coursework: Data Structures, Algorithms, Linear Algebra, Calculus, Probability, Statistics

EXPERIENCE

Full-Stack Software Engineer (Educational Platform)

Nov. 2021 – Jul. 2025

SMATHS-Kids Initiative – Ogun State, Nigeria

- Engineered and maintained a production learning platform (smathskids.com) serving 700+ active students
- Designed RESTful backend services and admin dashboards for grading, analytics, and student tracking workflows
- Implemented automated reporting and data pipelines, reducing manual administrative workload by 40%
- Improved system reliability and scalability during user growth through backend refactoring and workflow optimization

Software Engineer & Automation Developer

Jan. 2023 – Jul. 2025

Apt Scholars Universal College – Ogun State, Nigeria

- Built Python-based automation systems for quiz generation, grading pipelines, and student performance analytics
- Developed data processing workflows using Pandas and Excel automation, reducing grading turnaround time by 50%
- Improved instructional efficiency by replacing manual workflows with reusable scripting tools for educators
- Recognized as 2nd Best Teacher (Cowbellpedia STEM 2025); mentored students to national competition finals

Data Management Intern

Aug. 2021 – Dec. 2022

Smile Pride Dental Clinic – Ogun State, Nigeria

- Digitized and maintained confidential patient records with high accuracy across structured medical datasets
- Cleaned and corrected inconsistencies in legacy databases, improving data reliability for reporting workflows

PROJECTS

Pencil – Full-Stack Assessment Platform (Flagship)

React.js, Node.js, PostgreSQL, Redis, Docker

- Built and deployed a full-stack assessment platform supporting exam creation, automated grading, analytics, and role-based workflows
- Designed relational PostgreSQL schemas and REST APIs for authentication, user management, assessment delivery, and grading workflows
- Implemented Redis-backed caching and asynchronous processing to improve performance across high-frequency assessment operations
- Containerized application services with Docker and designed isolated deployment workflows for reliable development and production environments

Clueline – AI Error Management Platform

React.js, Next.js, Node.js, PostgreSQL, LLMs

- Built an AI-powered error management platform that captures application failures and translates technical errors into plain-English explanations for end users
- Developed an error ingestion pipeline capturing stack traces, user feedback, and session context to give developers actionable incident data
- Integrated LLM-powered root-cause analysis to generate developer-facing explanations and suggested fixes using application and repository context
- Built a developer dashboard for monitoring, grouping, investigating, and resolving application errors alongside AI-generated insights

DejaWHO – Cognitive Analytics Platform

React.js, Node.js, PostgreSQL, Machine Learning

- Built a gamified cognitive assessment platform for tracking memory, reasoning, and behavioral performance trends
- Integrated ML-based scoring models for personalized cognitive performance analysis over time
- Designed secure backend data architecture for storing longitudinal user session data

LEADERSHIP & INVOLVEMENT

- Member, Pearson Scholars Social Planning Committee – Organized campus-wide community events
- Community Advocate – Delivered STEM outreach and youth empowerment presentations

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, SQL, C
- Frontend:** React.js, Tailwind CSS
- Backend:** Node.js, Express.js, FastAPI
- Databases & Tools:** PostgreSQL, Redis, Docker, Git, REST APIs
- ML/Data:** Scikit-learn, Pandas, NumPy